BILAL AHMED

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EDUCATION

Habib University, Karachi Jun 2026

BSc Computer Science **CGPA: 3.24**

In Preparation: *AZ-900, SC-900, MD-102, SC-300, CompTIA Security+*

WORK EXPERIENCE

Project Trainee - Microsoft | JBS Consulting Aug 2026 – Present

- Completed a structured **Microsoft 365/Entra ID** onboarding, building a demo tenant covering identity management, licensing, **RBAC**, and **Conditional Access**.
- Designed and validated location-based **Conditional Access** policies (trusted-IP, country-based, geo-exception) in a training environment, using report-only testing to confirm behavior before simulated enforcement.
- Practiced **Microsoft Graph PowerShell** and **Exchange Online PowerShell** for user provisioning, licensing, and **RBAC** validation across a ~20-user test environment and implemented **Privileged Identity Management (PIM)** for time-bound role assignments.

Intern - Information Security | Mari Energies Jul 2026 – Jul 2026

- Ran **Nessus vulnerability assessments** (credentialed & non-credentialed) across Ubuntu, Windows Server, and Windows 10 hosts, cutting Windows 10 critical/high findings from **9 to 0** through patch remediation and rescanning.
- Mapped **ISO/IEC 27001:2022** Annex A's **93 controls** to scan findings for risk documentation, and analyzed critical **CVEs** (Windows TCP/IP, Print Spooler, MSHTML) for prioritized remediation.
- Built and delivered **GoPhish** phishing-simulation campaigns end-to-end (templates, tracked landing pages**, Mailpit/Gmail SMTP**, **Cloudflare Tunnel** delivery).
- Completed a one-week rotation with the **Network Security** team, gaining exposure to **leaf-spine** and **campus network** architecture, **VLAN**/switch/router configuration (**Cisco Packet Tracer**), **SD-WAN**, and **AAA/802.1x** authentication concepts.

Intern - Information Security | SECMC (Sindh Engro Coal Mining Company) Jul 2025 - Aug 2025

- Analyzed **SOC workflows** integrating **SIEM, SOAR, and EDR** for automated threat response; performed detailed **Wireshark** packet analysis to evaluate **TCP/IP** attacks and encrypted traffic.
- Conducted system audits using **Nessus**, mapped vulnerabilities to **CVE/CWE** databases for prioritization, and explored **OWASP Top 10** risks including **Injection** and **Broken Access Control**.
- Evaluated identity and access management (**IAM/PAM**) controls and mapped security requirements across **IaaS**, **PaaS**, and **SaaS** cloud models, with direct relevance to cloud endpoint and compliance frameworks.
- Compared **Symmetric/Asymmetric encryption** and hashing trade-offs to protect enterprise data confidentiality and integrity.

PROJECTS

AI-Driven Executive Chat & Reporting Portal | Alkhidmat: Oct 2025 - May 2026

- Built a full-stack conversational AI platform (React, Node.js, FastAPI) enabling natural-language queries over organizational data.
- Implemented a secure data access layer with JWT authentication, gateway-owned SQL execution, and RBAC policies; the sandboxed Python AI service generates but never directly executes queries.

Prompt Injection & MCP Exploits: A Cryptographic Provenance Perspective on LLM-Agent Security: Mar 2026 - May 2026

- Surveyed prompt-injection and **Model Context Protocol (MCP)** attack classes, mapping six MCP-specific vulnerability classes including **tool poisoning** and **tool shadowing**.
- Built a **LangChain + fastmcp** agent and reproduced four injection attacks across five defense configurations (**4×5 matrix**).
- Measured a **100% attack success rate** for undefended indirect prompt injection, reduced to **15%** under a secondary-model guardrail.
- Designed a **SHA-256 hash-pinned** tool-manifest defense, operationalizing a cryptographic provenance model analogous to **TLS/PKI**.

VPC Segmentation & NAT Egress Implementation: Oct 2025 - Dec 2025

- Architected a tiered **Virtual Private Cloud (VPC)** on AWS, moving from a publicly accessible flat network to a segmented environment with private subnets to reduce the attack surface.
- Engineered a custom Linux **NAT Router** using **iptables masquerading** and **kernel-level IP forwarding**, configuring VPC route tables to direct all 0.0.0.0/0 traffic to the NAT instance for secure, centrally monitored internet egress.
- Enforced granular **access controls** through **Security Groups** and **Network ACLs**, and performed baseline security audits using **ICMP**, **Traceroute**, and **HTTP timing**.

SKILLS

- **Security & Networking:** Wireshark, Nessus, Nmap, GoPhish, ISO/IEC 27001, VPC Configuration, NACLs, RBAC, OWASP Top 10, CVE/CWE Analysis
- **Microsoft Cloud & Identity:** Microsoft Entra ID, Conditional Access, Privileged Identity Management (PIM), Microsoft Graph PowerShell, Exchange Online, RBAC, M365 Administration
- **Cloud & Infrastructure:** AWS (IoT Core, S3, Athena, VPC), Microsoft Azure, Docker, Linux
- **Programming & Scripting:** Python (security scripting, automation), C++, SQL (MySQL, PostgreSQL, SQLite), Bash
- **AI & Data:** LangChain, RAG pipelines, Power BI, Pandas
- **Tools & Platforms:** Git, VirtualBox, Windows Server / Windows OS Administration